

INSTRUCTIONS TO THE CANDIDATES

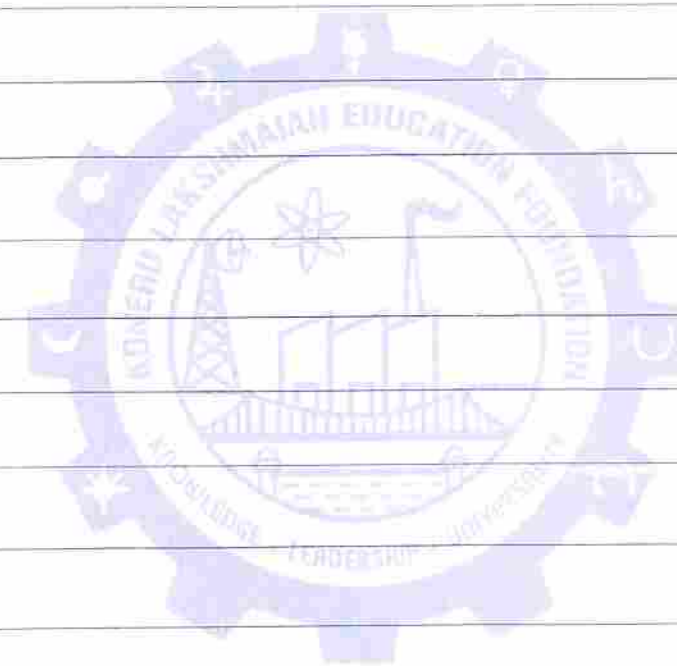
- ❖ Do not use sketch pens/ high –lighter pens/ color pencils etc., in underlining / highlighting certain points in the answer or while drawing sketches etc.,
- ❖ Check the Answer booklet thoroughly. The defective answer booklet may be returned to the invigilator before starting the examination.
- ❖ Ensure that the OMR Barcode sheet is properly secured to the answer booklet.
- ❖ Do not write on or tamper the Barcode. If you do so, your booklet will not be evaluated.
- ❖ The candidate is prohibited from writing his/ her NAME / ADDRESS/ PHONE NUMBER / EMAIL ID / Religion or any other matter on any part of the answer booklet / drawing sheet/ graph sheet/ addressing the examiner in any manner whatsoever in the answer booklet. The candidate is not supposed to tear a page in full or part of the booklet.

Non-adherence with this instruction shall result in candidate being booked under malpractice.

- ❖ Answer should be written on both sides of the paper. Candidate should write in all 25 lines in each page. It is not necessary to begin each answer in a fresh page.
- ❖ Candidate should write only question number in the margin.
- ❖ Candidate should return the answer booklet to the invigilator before leaving the examination hall.
- ❖ **ADDITIONAL SUPPLEMENT OF ANSWER SHEETS WILL NOT BE ISSUED.**

INSTRUCTIONS TO THE EXAMINERS

- ❖ Invigilators should ensure that the above instructions are strictly followed by the candidates.
- ❖ After verifying the particulars on the OMR sheet, invigilator should sign in the box specified.
- ❖ Do not crumple or cut the sides of the sheets.
- ❖ Do not write, mark or damage the timing tracks or Barcode.



3

4 A

A

Display KLV on LCD

ORG 0H

MOV A, #38H

ACALL COMNVRT

ACALL DELAY

MOV A, #040E

ACALL COMNVRT

ACALL DELAY

MOV A, #01

ACALL COMNVRT

ACALL DELAY

MOV A, #06

ACALL COMNVRT

ACALL DELAY

MOV A, #82

ACALL COMNVRT

~~MOV A, A~~ ACALL DELAY


```

MOV A, #'K'
ACALL DATAWRT
ACALL DELAY
MOV A, #'L'
ACALL DATAWRT
ACALL DELAY
MOV A, #'V'
ACALL DATAWRT
ACALL DELAY

```

COMNWRT:

~~DATA WRT~~

```

MOV P1, A
CLR P1.0
CLR P2.0
SETB P2.1

```

DATA WRT:

```

MOV P1, A
SETB P1.0
CLR P2.0
SETB P2.1

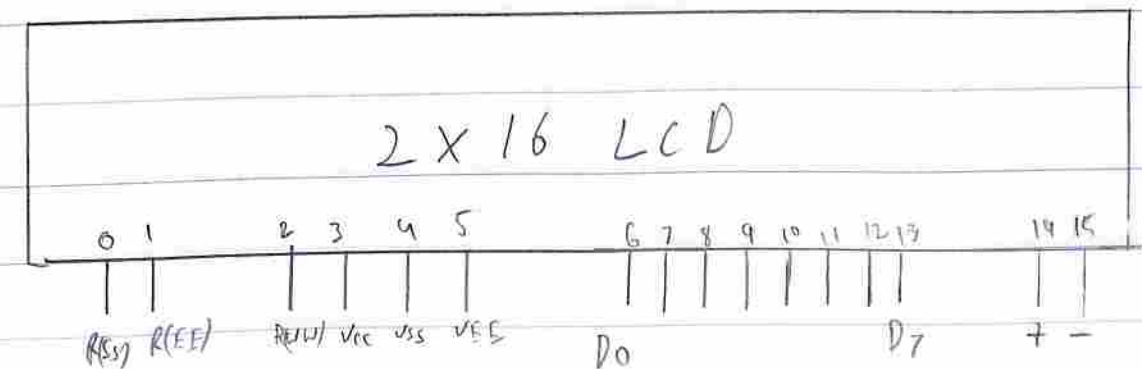
```

DELAY:

```
MOV R0, #107H.
```

END.

B.



LCD - Liquid Crystal Display

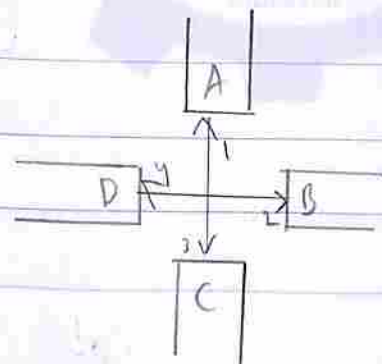
2 A

Stepper Motor

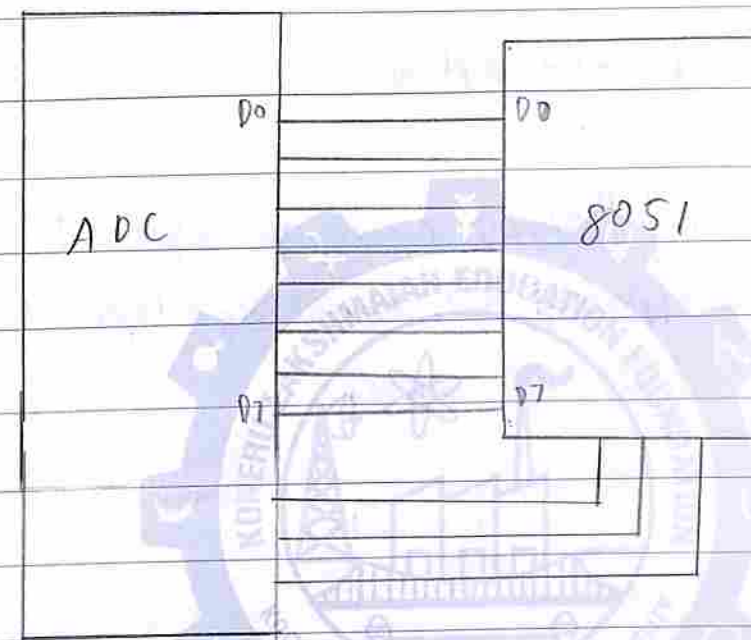
Wave Drive 4 Step

clockwise	step	A	B	C	D	Anticlockwise
	1	1	0	0	0	
	2	0	1	0	0	
	3	0	0	1	0	
	4	0	0	0	1	

clockwise



B



The ADC converts Analog signal to digital signals.

The ADC interfaces with 8051 by connecting with from D0 to D7.

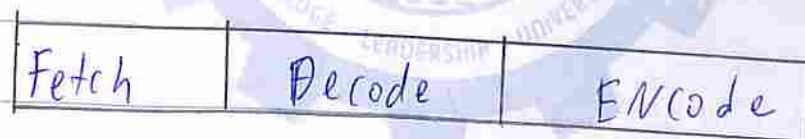
D

Arm Pipeline.

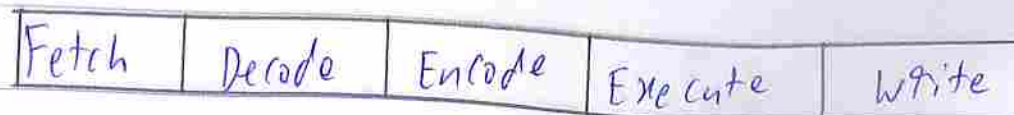
There are mainly 3 Arm pipelines.

Arm 7	Pipelining can be very helpful. It makes multi-processing fast, w/ simultaneously multiple tasks can do it's process.
Arm 9	
Arm 10	

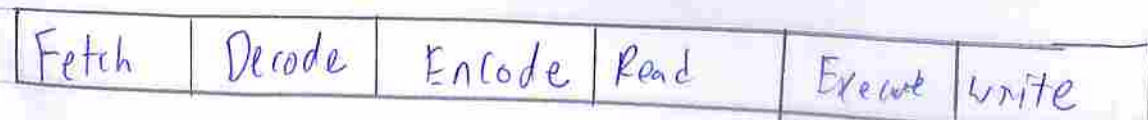
Arm 7 Backwards compactable



Arm 9



Arm 10



C

Arm processor mode.

There are 2 Arm processor mode.

- User mode
- Complex mode.

→ User mode

- Simple tasks
- Interacts with others.
- has small access to memory

→ complex mode

- complex tasks
- has full access to memory

1

A

We need 8 pins to interface the ADC0802 with the 8031 micro controller.

It connects from ~~D0~~ to D₇
D₀ to D₇

B

We can scroll text on 2x16 LCD using 8031 micro controller.

It will happen when they are both linked and connected. By passing ~~PR~~ V_{EE}.

LCD → Liquid Crystal Display

2x16 → 2 rows 16 columns.

Q. C

The DAC is used to convert Digital signals to analog signals.

It can be interfaced with 8051 micro controller.

It takes inputs from V_R.

D

PIC micro controller.

Peripheral Instruction controller.

It used to for peripheral instruction which are to be controlled.

F

The Timer module in the PIC micro controller.

It can be used as a clock. It can maybe used as a clock pulse, or a stopwatch, or

it maybe asynchronous or maybe synchronous. I

don't know. But the only thing I know is that this

course is not set on me, sorry if this was a mistake but

Thank you for having this paper as my platform of speech.

6

A

PIC

- It is a 16 bit Instruction set
- It follows Von-Neuman architecture
- Peripheral Instruction Controller,

B

Von-Neuman	Harvard
* Now architecture follows this	* This has become out dated
- All memory is individual.	* All memory is split into parts
* Fast and efficient	* slow as Ram is shared

